Two flavors of control in Georgian: *decide* and *try*[*]

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1. Introduction

Across a variety of languages, there are certain clause-embedding predicates which license *obligatory control* (henceforth OC) constructions, whereby the subject of a complement clause must be bound by a DP argument in the clause which immediately dominates it. Landau (2013) proposes that such constructions display the **OC signature** in (1).

(1) *The OC signature*

A construction $[\ldots X_i \ldots [_S\, Y_i \ldots] \ldots]$, where Y is the subject of embedded clause S:

a. X must be (a) co-dependent(s) of S, AND

b. Y (or part of it) must be interpreted as a bound variable.
(X = controller; Y = controllee)

Traditionally, OC predicates have been classified into two different kinds: **partial control** and **exhaustive control** (e.g., Landau 2000). The central property which distinguishes these two kinds is that for partial control, the controllee may be interpreted as *semantically plural*, denoting some non-atomic entity of which the controller is a proper part. Conversely, exhaustive control requires strict identity between the controller and controllee.

Until now, it has been largely assumed that which category an OC predicate belongs to is fixed. In this paper, I will challenge this view by comparing two control predicates in Georgian, *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying.' What makes Georgian (Kartvelian) a language worth examining is it has two different ways of licensing control constructions syntactically: (i) finite clauses with subjunctive morphology, and (ii) nominalized predicates, often called *masdars* (e.g., Aronson 1990). Both strategies for OC are licensed by *decide* and *try* in Georgian, but their profile differs with respect to the choice

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of syntactic complement: whereas *gadac'qvet's* 'he/she will decide' permits partial control (henceforth PC) with both finite subjunctive clauses and masdars, *cdilobs* 'he/she is trying' licenses PC with finite subjunctive clauses and requires exhaustive control (henceforth EC) with masdars. The contrast between these two control predicates is summarized in Table 1.

Table 1: Types of OC with Georgian control predicates

	Finite subjunctive clause	*Masdar* (nominalized predicate)
gada-c'qv-et'-s *decide*	PC	PC
cdil-ob-s *try*	**PC**	EC

Although *decide* seems to behave as one would expect for PC predicates, *try* presents the more interesting case. It is generally regarded as an EC predicate (Landau 2000, White and Grano 2014, Pearson 2016, *i.a.*), so the alternation between EC for masdars and PC for finite subjunctive clauses is surprising and will constitute the central puzzle of this paper.[1]

My analysis will involve three key ingredients:

- there is a structural difference (besides nominalization) between non-finite masdars and finite subjunctive clauses in control constructions, reflected in the size of their embedded clause and, more importantly, whether their tense feature is valued;
- in terms of selectional restrictions, *decide* is future-oriented and requires its complement to structurally bear future/irrealis TMA, whereas *try* has no such restriction and naturally accepts tenseless complement clauses;
- according to the *synthesis* model of complementation (Wurmbrand and Lohninger 2023), syntax is autonomous (free merge) and builds up structure freely, so long as the result satisfies the semantic requirements of the clause-embedding predicate.

2. *Decide* and *try* are OC predicates in Georgian

Before we proceed, it is important to establish that *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' are indeed OC predicates. For a control construction to display the OC signature (1), the controller X and controllee Y are required to stand in a particular syntactic relationship: Y receives a bound variable interpretation, and therefore must be in the *c*-command domain of X. Conversely, any configuration where X does not *c*-command Y is predicted to be ungrammatical, as shown in (2) for English.

(2) [Bill$_i$'s friends]$_j$ wished [PRO$_{j/*i}$ to congratulate {themselves$_j$/*himself$_i$}].

[1] Wurmbrand (1998) notes that in German, there is a marked variant of *try* which is unlike the standard variant in permitting irrealis (rather than untensed) interpretations of the complement clause. This variant is incompatible with restructuring properties like long distance scrambling and marginally allows PC; Georgian *try* is similar in the sense of being flexible with respect to the PC/EC distinction, though it is highly systematic with regards to the syntactic category of the complement and the kind of OC it triggers.

Moreover, the relationship between controller and controllee is subject to certain locality restrictions: although the two constituents are non-clause mates, they cannot be separated by an intervening clause. The impossibility of long-distance control across multiple clause boundaries in OC constructions is illustrated in (3), where the local subject *Mary* of control predicate *realize* cannot be co-indexed with PRO in the deeply embedded clause, unlike overt pronominal subjects which are unrestricted in their choice of antecedent.

(3) a. ***Mary**$_i$ realized that John hated [**PRO**$_i$ to nominate herself]. (Landau 2024)
b. **Mary**$_i$ realized that John would hate [for **her**$_i$ to nominate herself].

Another diagnostic for the locality of OC is that the interpretation of the controllee PRO is restricted to only sloppy readings in the complement clause of elided VPs; pragmatic co-reference yielding the strict reading is blocked, as shown in (4).

(4) Ted$_i$ promised [PRO$_i$ to attend the commencement ceremony], and Sue$_j$ did too ~~promise [PRO$_{j/*i}$ to attend the commencement ceremony]~~.

The data in (5)-(8) below illustrate that Georgian clause-embedding predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' display characteristics of the OC signature with both masdar complements and finite subjunctive complement clauses.

Obligatory sloppy readings of PRO with masdars

(5) šota-m **gada-c'qv-it'-a** [PRO c'ign-is c'a-k'itx-va], maia-ma-c
Shota-ERG **PVB-decide-AOR.3SG** book-GEN PVB-read-NMN Maia-ERG-too
'Shota$_i$ decided [PRO$_i$ to read a book], and so did Maia$_j$ ~~decide [PRO$_{j/*i}$ to read a book]~~.'

(6) mariam-i **cdil-ob-d-a** [PRO k'ar-is ga-ǧ-eb-a]-s, otar-i-c
Mariam-NOM **try-TS-IMPF-3SG** door-GEN PVB-open.NMN-DAT Otar-NOM-too
'Mariam$_i$ was trying [PRO$_i$ to open the door], and so was Otar$_j$
~~trying [PRO$_{j/*i}$ to open the door]~~.'

Non-c-commanding control disallowed with PRO in finite subjunctive clauses

(7) [šota-s$_i$ deda-m]$_j$ **gada-c'qv-it'-a**, [rom {PRO$_{*i/j/*k}$/mas$_{i/j/k}$}
Shota-GEN mother-ERG **PVB-decide-AOR.3SG** that {PRO/3SG.DAT}
c'ign-i c'a-e-k'itx-a]
book-NOM PVB-PV-read-PLUPERF.3SG
'[Shota$_i$'s mother]$_j$ decided {PRO$_j$ to read a book / that someone$_{i/j/k}$ reads a book}.'

(8) [maia-s$_i$ megobar-i]$_j$ **cdil-ob-d-a**, [rom {PRO$_{*i/j/*k}$/mas$_{i/j/k}$}
Maia-GEN friend-NOM **try-TS-IMPF-3SG** that {PRO/3SG.DAT}
šejibr-i mo-e-g-o]
competition-NOM PVB-PV-win-PLUPERF.3SG
'[Maia$_i$'s friend]$_j$ was trying {PRO$_j$ / for someone$_{i/j/k}$} to win the competition.'

Control in finite complement clauses has been attested cross-linguistically, notably Balkan languages (e.g., Aromanian), as well as other languages like Amharic and South Saami — see Landau (2024) for more discussion on finite control. Georgian likewise permits finite clauses to participate in OC constructions, as in (7) and (8), under two conditions:

Necessary conditions for finite control in Georgian

1. the embedded predicate has *subjunctive morphology* (e.g., pluperfect);
2. the subject of the embedded clause is null PRO.

If the embedded predicate had instead been inflected for the indicative mood (without loss of grammaticality), then no control would obtain, as shown with the indicative conditional complement of *gadac'qvet's* 'he/she will decide' in (9), where the null subject could refer to Shota, his student Maia, or to some other contextually salient individual in the discourse.

(9) šota$_i$ maia-s$_j$ masc'avlebel-i-a da man$_i$ gada-c'qv-it'-a, [rom
Shota Maia-DAT teacher-NOM-COP and he.ERG PVB-decide-AOR-3SG that
pro$_{i/j/k}$ c'ign-s ca'-i-k'itx-av-d-a]
book-DAT PVB-PV-read-TS-IMPF-3SG
'Shota$_i$ is Maia$_j$'s teacher, and he$_i$ decided that someone$_{i/j/k}$ would read a book.'

The null embedded subject in (9) thus exhibits the behavior of *pro*. Georgian is a *pro*-drop language, which raises the question of whether one can similarly treat the covert subject in (7) and (8) as being instances of *pro* with accidental co-reference to the matrix controller. This cannot be the case, however, since it differs from the overt subject pronoun *mas* with respect to binding; in particular, the covert DP *must* be co-indexed with the matrix controller, even when other individuals are discourse-salient. Given that (7) and (8) display characteristics of OC, we conclude that the subject of the embedded clause is null PRO.

Taking stock of what we have seen so far, the Georgian predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' license OC with both nominalized complements (masdars) and finite subjunctive clauses with null subjects. In the next section, I will discuss how they differ with respect to the kind of OC triggered by their syntactic complement.

3. Partial vs. exhaustive control

In a control construction, PC and EC differ with respect to whether the controllee may be interpreted as including the controller or requiring it to supply exhaustive reference. One common way of determining the type of OC involved is to embed a *collective predicate* under a control verb whose subject is semantically singular; as shown in (10), the collective predicate *gather* is compatible with OC constructions embedded under *decide*, but not *try*:

(10) a. The chair$_i$ decided [PRO$_{i+}$ to gather during the strike].
b. *Kim$_i$ tried [PRO$_i$ to gather at noon].

Another difference between PC and EC is in their ability to tolerate temporal mismatches between the matrix and embedded eventualities. This contrast is reflected more clearly by the placement of temporal adverbials in their respective clauses: whereas *decide* is future-oriented and permits the embedded reference time to be posterior to the matrix reference time, *try* requires the matrix and embedded reference times to be strictly overlapping, and thus cannot accommodate cases where the two time intervals are temporally disjoint (11).

(11) a. **Yesterday**, John decided [PRO to leave **tomorrow**]. (Wurmbrand 2014)
b. **Yesterday**, John tried [PRO to leave (***tomorrow**)].

The connection between tense and the kind of OC established was previously identified in Landau (2000, 2004); in particular, Landau notes that PC corresponds to complements that involve tense, whereas EC corresponds to complements which are semantically untensed. He argues that despite being tenseless, the complement clause of EC constructions still maintains a TP projection in the syntax, but whose tense feature is deficient. Wurmbrand (2014) further suggests that for non-finite complement clauses that license PC, e.g. certain English infinitives, the semantic "tense" which is detected in the embedded predicate comes from another aspectual/modal element in the TMA domain. These ideas will be useful for the analysis that I propose in the next section, but for now, it suffices to keep in mind the observation that PC-complements are tensed, while EC-complements are untensed.

Turning to the Georgian control predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying,' a similar pattern emerges with non-finite masdar complements: in (12), *gadac'qvet's* 'he/she will decide' can include the collectivizer *ertad* 'together' inside the embedded clause, but not *cdilobs* 'he/she is trying,' to indicate a semantically plural PRO.

Georgian control with collective predicate in masdar complements

(12) a. šota-m$_i$ gada-c'qv-it'-a [PRO$_{i+}$ **ertad** saseirno-d c'a-svl-a]
Shota-ERG decided.3SG **together** walking-ADV PVB-go-NMN
'Shota decided to go walking together.'
b. šota$_i$ cdil-ob-d-a [PRO$_i$ (***ertad)** saseirno-d c'a-svl-a-s]
Shota try-TS-IMPF-3SG **together** walking-ADV PVB-go-NMN-DAT
'Shota was trying to go for a walk (*together).'

Moreover, a temporal mismatch is possible between *gadac'qvet's* 'he/she will decide' and its masdar complement, but not *cdilobs* 'he/she is trying' (13). Together, these data support that a PC vs. EC distinction is at play in nominalized complements of the two predicates.

Temporal mismatch between Georgian control predicates and masdar complements

(13) a. **gušin** šota-m gada-c'q'v-it'-a [PRO c'ign-is **xval** c'a-k'itx-va]
yesterday Shota-ERG PVB-decide-AOR-3SG book-GEN **tomorrow** PVB-read-NMN.NOM
'Yesterday, Shota decided to read a book tomorrow.'

b. **gušin** šota e-cad-a [PRO c'ign-is **(*xval)** c'a-k'itx-va]-s
yesterday Shota PV-try-AOR.3SG book-GEN PVB-read-NMN-DAT
'Yesterday, Shota tried to read a book (*tomorrow).'

When the control predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' embed finite subjunctive clauses, however, no contrast is observed with respect to including the collectivizer *ertad* 'together' in the complement. Both of them allow the covert PRO in the embedded subject to be semantically plural (14), suggesting that they trigger PC.

Georgian control predicates with collective predicate in finite subjunctive clauses

(14) a. šota-m gada-c'qv-it'-a, [rom PRO saseirno-d **ertad**
Shota-ERG PV-try-AOR.3SG that walking-ADV **together**
c'a-sul-i-qv-nen]
PVB-go-TS-be.PLUPERF-3PL
'Shota decided to go for a walk together.'

b. šota cdil-ob-d-a, [rom PRO saseirno-d **ertad**
Shota try-TS-IMPF-3SG that walking-ADV **together**
c'a-sul-i-qv-nen]
PVB-go-TS-be.PLUPERF-3PL
'Shota tried to go for a walk together.'

Thus, the Georgian control predicate *cdilobs* 'he/she is trying' seems to alternate between EC and PC, depending on whether its complement is a masdar or a finite subjunctive clause. The data in (12)-(14) confirm the OC generalization stated in Table 1. In the next section, I will present my proposal for how to explain these facts.

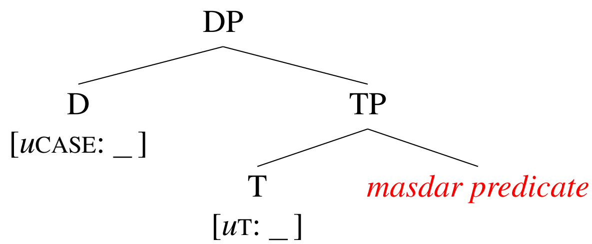
4. Syntactic analysis

In terms of finiteness, there is a clear separation between masdars and subjunctive clauses: only the latter requires that its verb be inflected for tense and mood.[2] In Georgian, I claim that the key defining property for finiteness is *tense*, specifically, whether T° in the extended verbal projection has a valued tense feature (see Huang 2022 and He 2024 for proposals of a similar vein in Mandarin Chinese). If this functional head is not independently valued for tense, then the whole verbal phrase must be restructured into a masdar via nominalization; otherwise, it will maintain the status of a tensed clause with inflectional agreement on the verb.

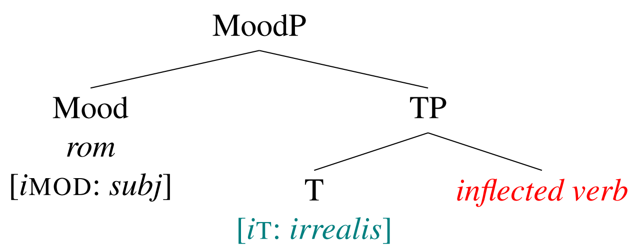
The relevant generalizations are stated in (15) and (16) for masdars and finite embedded clauses, respectively, in conjunction with their syntactic representations.

[2] In control constructions, masdars can also be sensitive to viewpoint aspect, which provides empirical support that they minimally project AspP. For masdars, like with verbal roots, (im)perfective aspect is often indicated by the presence/absence of an aspectual preverb (e.g., Aronson 1990:50, fn. 6).

(15) **Generalization #1:** Masdar complements only project up to TP in their extended verbal projection (i.e., the layers below their surface nominal form). Despite being syntactically realized as TPs, the tense feature on their T head must be unvalued.[3]



(16) **Generalization #2:** Finite embedded clauses may project up to CP; however, truncation is also possible, so long as the result is not any smaller than TP and has valued tense. In particular, subjunctive clauses are MoodP marked by *irrealis* tense.



By *irrealis* tense, I am not referring to an absolute/indexical tense like PRES or PAST, which serve to anchor the reference time, but rather appeal to the notion of a relative tense which specifies the relation between the matrix and embedded reference times, namely the latter being posterior to the former. Landau (2004) also refers to this as *dependent tense*.

In addition, I consider the Georgian control predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' to place distinct semantic requirements on their complement. On one hand, *decide* requires the embedded eventuality to be temporally posterior to the matrix event; on the other hand, *try* is naturally compatible with embedded eventualities which are interpreted as simultaneous to the matrix event. These semantic requirements are a reflex of their selectional properties: within narrow syntax, *decide* selects complement clauses that structurally bear future/irrealis TMA, which is subsequently interpreted at the interface with LF; besides its complement being non-propositional, *try* does not have such a selectional restriction.

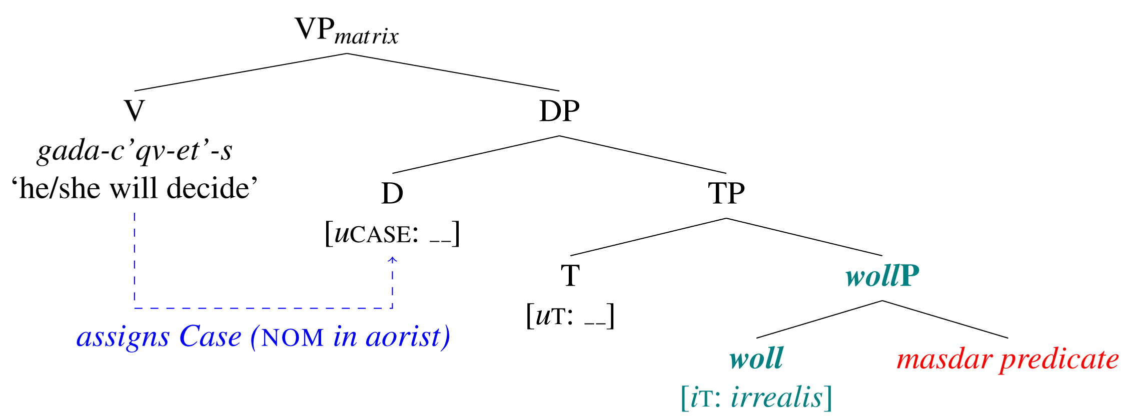
With this framework in mind, let us first consider how control predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' interact with finite subjunctive clauses, whose syntactic category is MoodP with *irrealis* tense. Bearing the necessary tense feature, this class of complements naturally conforms to the semantic requirements of *decide*, and are also compatible with *try*. Moreover, since finite clauses are tensed, they are correctly predicted to allow PC for both control predicates, as corroborated in (14). So far, so good.

[3] An *n*P-phrase is also projected below DP by the nominal categorizing head, which selects TP with null PRO in its specifier for the control complement; these nodes have been omitted for clarity in presentation.

What about masdars? These nominalized complements come with an unvalued tense feature on embedded T°, which typically require them to receive tense specification from matrix T° via Feature Valuation and Agree (Chomsky 2000), yielding a simultaneous reading for the matrix and embedded events. Given that they are tenseless, masdars are thus compatible with *cdilobs* 'he/she is trying' and will trigger EC (12b). But if *gadac'qvet's* 'he/she will decide' selects complements that are marked for future/irrealis TMA, why is it compatible with masdars, and why do such control constructions allow PC (12a)?

I argue that this mismatch can be reconciled by the *synthesis* model of complementation (Wurmbrand and Lohninger 2023), which assumes the matrix and embedded predicates to interact *dynamically* with each other in the derivation, and that syntax builds up structure freely so long as the result will satisfy the semantic requirements of the embedding verb. Since *gada-c'qv-et's* 'he/she will decide' needs future/irrealis TMA on its complement, the syntax projects a covert modal *woll* below the TP nucleus of the masdar (cf. Wurmbrand 2014), prior to nominalization, as a way of fulfilling this selectional requirement.

(17)



Thus, even in non-finite masdars, the complement bears irrealis TMA and will trigger PC. If masdars can introduce irrealis tense by means of a covert future modal *woll*, why doesn't *cdilobs* 'he/she is trying' also project *woll*P in its nominalized complements? I claim that the reason is twofold: first, there is a principle of economy at stake. The predicate *cdilobs* 'he/she is trying' does not impose selectional restrictions for TMA on its complement; consequently, the masdar does not require any additional structure, and merging *woll* into the syntactic derivation would be costly. Second, there is pragmatic competition: *cdilobs* 'he/she is trying' has a clear manner of expressing future/irrealis complements with finite subjunctive clauses, and a clear manner of expressing tenseless complements with masdars. From a processing standpoint, permitting masdars to freely project *woll*P would therefore be strongly dispreferred — not only would it create an ambiguity with respect to whether the masdar is semantically tenseless/tensed, but the derived meaning with the covert modal would already be expressible by other means, namely with a finite subjunctive clause.

To summarize: the control predicate *gadac'qvet's* 'he/she will decide,' by virtue of its selectional restrictions for future/irrealis TMA, always permits PC on its complement, whether a finite subjunctive clause or a masdar. By contrast, the control predicate *cdilobs* 'he/she is trying' is compatible with both finite subjunctive clauses that have irrealis tense, and with non-finite masdars that are untensed. As a result, it exhibits alternation between PC and EC depending on the syntactic nature of its complement.

5. Conclusion and Extensions

This paper has shown that the Georgian control predicates *gadac'qvet's* 'he/she will decide' and *cdilobs* 'he/she is trying' exhibit different patterns in the kind of OC that they trigger. Whereas *decide* permits PC with both finite subjunctive clauses and masdars, *try* requires EC with masdars but does allow PC in finite subjunctives as well.

The key to this puzzle, I argue, lies in the fact that *decide* places a selectional restriction that its complement structurally bear irrealis TMA, which masdar complements respond to by projecting *woll*P in their syntactic derivation. This additional merge operation satisfies the semantic requirements of the future-oriented *decide*, and thus constitutes a valid output of the grammar according to the *synthesis* model of complementation (Wurmbrand and Lohninger 2023). The predicate *try* does not restrict its complement in such a manner, maintaining non-finite masdars as tenseless while still remaining compatible with finite subjunctive clauses that have irrealis tense.

The ability for *cdilobs* 'he/she is trying' to alternate between PC and EC suggests that the kind of OC should not be treated as a static property of control predicates, but rather may depend on the syntactic category of their complement. This dependency between the embedding predicate and its complement clause is also supported by the *synthesis* model, showing that the two interact dynamically in the syntactic derivation.

Landau (2000) and Wurmbrand and Lohninger (2023) have previously noted the unique behavior of *try*-complements, namely their semantic realization as both irrealis *and* tenseless. The connection between PC/EC alternation and syntactic category for complements of *cdilobs* 'he/she is trying' in Georgian could thus be explained from a slightly different perspective: finite subjunctive complement clauses are selected to bring out the *irrealis* meaning of the embedded event, whereas non-finite masdars are selected to bring out the tenseless interpretation, yielding simultaneity in the matrix and embedded events.

Although *cdilobs* 'he/she is trying' allows PC with finite subjunctive clauses, as shown by the availability of collective interpretations for the embedded predicate in (14), it does not allow a temporal mismatch between the matrix and embedded events like *gadac'qvet's* 'he/she will decide' does (18), supporting the absence of semantic tense in *try*-complements:

(18) a. gušin šota-m gada-c'qv-it'-a, [rom c'ign-i **xval** c'a-e-k'itx-a]
yesterday Shota-ERG decide.AOR.3SG that book-NOM **tomorrow** PVB-PV-read-PLUPERF.3SG
'Yesterday, Shota decided to read the book tomorrow.'

b. gušin šota e-cad-a, [rom c'ign-i **(*xval)** c'a-e-k'itx-a]
yesterday Shota tried that book-NOM PVB-read-PLUPERF.3SG
'Yesterday, Shota tried to read the book (*tomorrow).'

Thus, the PC/EC dichotomy does not perfectly correspond to whether a temporal mismatch is permissible. As Landau (2015) and Pearson (2016) have argued, the notion of semantic tense that distinguishes PC from EC may be derived from the more fundamental distinction between complements of attitude and non-attitude predicates, with *try* being a special case of the former which accepts tenseless complements. Extending the analysis in Georgian for *decide*, *try*, and other control predicates along these lines may also prove to be fruitful.

References

Aronson, Howard Isaac. 1990. *Georgian: A Reading Grammar*. Slavica Publishers.

Chomsky, Noam. 2000. Minimalist Inquiries: The Framework. In *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik*, 89–155. Cambridge, MA: MIT Press.

He, Yuyin. 2024. Finiteness in Mandarin clausal complements: The role of ICH and future modals. *Journal of East Asian Linguistics* 33:259–297.

Huang, C.-T. James. 2022. Finiteness, opacity, and Chinese clausal architecture. In *New Explorations in Chinese Theoretical Syntax: Studies in honor of Yen-Hui Audrey Li*, ed. by Andrew Simpson, Linguistik Aktuell/Linguistics Today, 17–76. John Benjamins Publishing Company.

Landau, Idan. 2000. *Elements of Control: Structure and Meaning in Infinitival Constructions*. Dordrecht: Kluwer Academic Publishers.

Landau, Idan. 2004. The scale of finiteness and the calculus of control. *Natural Language & Linguistic Theory* 22:811–877.

Landau, Idan. 2013. *Control in Generative Grammar: A Research Companion*. Cambridge University Press.

Landau, Idan. 2015. *A Two-Tiered Theory of Control*. MIT Press.

Landau, Idan. 2024. Noncanonical obligatory control. *Language and Linguistics Compass* 18:1–43.

Pearson, Hazel. 2016. The semantics of partial control. *Natural Language & Linguistic Theory* 34:691–738.

White, Aaron Steven, and Thomas Grano. 2014. An experimental investigation of partial control. *Proceedings of Sinn und Bedeutung* 18:469–486.

Wurmbrand, Susi. 1998. Infinitives. Doctoral dissertation, MIT.

Wurmbrand, Susi. 2014. Tense and aspect in English infinitives. *Linguistic Inquiry* 45:403–447.

Wurmbrand, Susi, and Magdalena Lohninger. 2023. An implicational universal in complementation: Theoretical insights and empirical progress. In *Propositional arguments in cross-linguistic research: Theoretical and empirical issues, eds. Julia Hartmann and Angelika Wöllstein*, 183–232. Berlin: Mouton de Gruyter.

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